

Chapter 8

Arithmetic Geometric

Sequences and Series

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

1) What is the 137th term of the sequence?
3, 6, 9, 12, 15, 18 ...

- A) 346 B) 401 C) 279 D) 472 E) 411

2) What is the sum of all multiples of 4 less than or equal to 400?

- A) 986 B) 3524 C) 1234 D) 8600 E) 20200

3) What is the next term in the sequence?
5, 7, 9, 11, 13, 15...

- A) 13 B) 17 C) 15 D) 25 E) 35

4) If the n th term of an Arithmetic Sequence is $4.5 - .5n$, what is the sum of the first four terms of the sequence?

- A) 13 B) 17 C) -15 D) -8 E) 2.5

5) In the following sequence, what is the 6th term?
 $9, 3, 1, \frac{1}{3}, \dots$

- A) $\frac{1}{27}$ B) $\frac{1}{81}$ C) $\frac{1}{9}$ D) 9 E) 27

6) What is the sum of the 89th, 99th, and 109th terms of the sequence?
5, 9, 13, 17...

- A) 505 B) 1297 C) 287 D) 1191 E) 441

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

7) If the 1st term of an arithmetic sequence is 8 and the 23rd term is 140, what is the **53rd term?**

- A) 320 B) 240 C) 509 D) 1120 E) 84

8) What is the sum of the **first 100 positive integers?**

- A) 1001 B) 801 C) 1901 D) 2501 E) 5050

9) What is the sum of all positive multiples of 5 **less than or equal to 500?**

- A) 98500 B) 15000 C) 1235 D) 8600 E) 25250

10) If the 14th term of an arithmetic sequence is 22 and the 83rd term is 298, **what is the 5th term?**

- A) -14 B) 8 C) 2.5 D) 4.5 E) -4

11) What is the 55th term of the geometric sequence; **$a, act, ac^2t^2 \dots$** ?

- A) $55ac^2t^2$ B) $a^{55}c^{55}t^{55}$ C) $ac^{55}t^{55}$ D) $ac^{54}t^{54}$ E) $a^{53}c^{54}t^{54}$

12) If the 5th term of an arithmetic sequence is 22 and the 55th term is 172, **what is the 105th term?**

- A) 652 B) 322 C) 282 D) 1290 E) 1492

13) What is the **177th term** of the sequence?
2, 4, 6, 8, 10...

- A) 354 B) 345 C) 435 D) 543 E) 45

Solutions Chapter 8

Arithmetic Geometric Sequences & Series

- | | | |
|------|-------|-------|
| 1. E | 6. D | 11. D |
| 2. E | 7. A | 12. B |
| 3. B | 8. E | 13. A |
| 4. A | 9. E | |
| 5. A | 10. A | |

1) In an arithmetic sequence, the same number is added or subtracted from the previous term to form the sequence. The number added, d , is called the common difference. To find the n^{th} term of an arithmetic sequence, use the formula: $A_n = A_1 + (n - 1)d$, where n is the number of terms, A_1 is the first term, and d is the common difference. To find the 137th term of the sequence: The first term is 3, and the common difference is 3 because 3 is added to each number in the sequence to get the next number.

$$A_{137} = 3 + (137 - 1)3 = 3 + (136)3 = 411$$

411 Answer E Level 5

2) Writing out a sequence or part of a sequence is usually a good plan when solving these questions where the sequence is not written out. All the multiples of 4 less than 400 would take way too long to write, so just write the first 3 or 4 terms to analyze the sequence: 4, 8, 12, 16 ... Notice this is an arithmetic sequence with a common difference of 4, and the first term is 4. The sum of a sequence is called a Series, and this question asks for the Sum. One equation for the Sum of an arithmetic sequence: $S_n = \frac{n}{2}(A_1 + A_n)$, where A_1 is the first term, and A_n is the n^{th} term or last term of the sequence. $A_1 = 4$, $A_n = 400$ but n is not known. To find the value of n , use the arithmetic sequence formula for the n^{th} term. $A_n = A_1 + (n - 1)d$ $400 = 4 + (n - 1)4 = 4n = 400, n = 100$. Many test-takers realized there were 100 terms without using the equation. Now use the formula to find the sum: $S_{100} = \frac{100}{2}(4 + 400) = 50(404) = 20200$.

20,200 Answer E Level 8

3) Just follow the pattern. Each term increases by two. The next term is 17, which is $15 + 2$.

17 Answer B Level 1

4) To find the terms in the sequence, start with 1 for the first term, 2 for the second term, 3 for the third term, and finally 4 for the last term and add the terms together. Use the given explicit equation to find the terms: $4.5 - .5n$, $n = 1, 2, 3, 4$. $(4.5 - .5(1)) + (4.5 - .5(2)) + (4.5 - .5(3)) + (4.5 - .5(4)) = 4 + 3.5 + 3 + 2.5 = 13$

13 Answer A Level 4

5) The sequence is a geometric sequence because each term is multiplied by the last term. The number each term is multiplied by is the common ratio, which can be found by taking any term in the sequence and dividing it by the term before it. In this sequence, the common ratio is $3/9$ or $1/3$. Since the first four terms are given, multiply the last term; $1/3$ by $1/3$ to find the 5th term and continue the same process for the 6th term $\frac{1}{3} \times \frac{1}{3} = \frac{1}{9} \times \frac{1}{3} = \frac{1}{27}$.

$\frac{1}{27}$ Answer A Level 3.5

6) The sequence is arithmetic because each term increases by four. To find the 89th, 99th, and 109th terms, use the formula for the n^{th} term of an arithmetic sequence: $A_n = A_1 + (n - 1)d$, where the first term A_1 is 5 and the common difference, $d = 4$. $A_{89} = 5 + (89 - 1)4 = 5 + 352 = 357$, $A_{99} = 5 + (99 - 1)4 = 5 + 392 = 397$,

$A_{109} = 5 + (109 - 1)4 = 5 + 432 = 437$. The sum of the 89th, 99th, and 109th terms: $357 + 397 + 437 = 1191$

1191 Answer D Level 6.5

7) To find the 53rd term of the arithmetic sequence, find the common difference using the n^{th} term of an arithmetic sequence formula: $A_n = A_1 + (n - 1)d$. $140 = 8 + (23 - 1)d$, $140 = 8 + 22d$, $d = 6$. Now use the formula again to find the 53rd term of the sequence.

$$A_{53} = 8 + (53 - 1)6 = 320$$

320 Answer A Level 6.5

8) To find the sum of the first 100 integers, use the sum of an arithmetic sequence formula (Arithmetic Series). When written out, the sequence is 1, 2, 3, 4... so it is an arithmetic sequence with a common difference of 1. The first term is 1, and the last term is 100, and there are 100 terms in the sequence. $S_{100} = \frac{100}{2}(1 + 100) = 5050$.

5050 Answer E Level 6

9) Writing out a sequence or part of a sequence is usually a good plan when answering these questions where the sequence is not written out. The sequence is 5, 10, 15, 20... which is an arithmetic sequence with the first term 5, the last term 500. Use the n^{th} term of an arithmetic sequence formula to find the number of terms: $A_n = A_1 + (n - 1)d$,
 $500 = 5 + (n - 1)5$, $500 = 5n$, $n = 100$. Now it seems obvious but use the formula if necessary. Finally, use a formula for the sum of an arithmetic sequence: $S_n = \frac{n}{2}(A_1 + A_n)$,

$$S_n = \frac{100}{2}(5 + 500) = 50(505) = 25250$$

25,250 Answer E Level 7

10) To solve this problem, set up two equations using the n^{th} term of an arithmetic sequence formula: $A_n = A_1 + (n - 1)d$. $22 = A_1 + (14 - 1)d$ or $22 = A_1 + (13)d$, the second equation from the given information $298 = A_1 + (83 - 1)d$ or $298 = A_1 + (82)d$. When given two equations with two variables, use substitution or elimination to solve. Do not graph because it takes too long. Align the equations and use elimination.

$$\begin{array}{r} A_1 + 13d = 22 \\ A_1 + 82d = 298 \end{array}$$

Multiply the first row by -1 , then add it to the second row so $69d = 276$, $d = 4$. Substitute 4 into either original equation to solve for A_1 : $A_1 + 13d = 22$. $A_1 + 13(4) = 22$, $A_1 = -30$. Using -30 as the first term and a common difference, $d = 4$, and the formula for the n^{th} term of an arithmetic sequence;

the 5th term of the sequence is $A_5 = -30 + (5 - 1)4 = -30 + 16 = -14$.

-14 Answer A Level 8.5

11) The sequence is geometric because each term is multiplied by ct to get to the next term. Remember, when multiplying by the same variable, add the exponents. Since the third term is ac^2t^2 , the fourth term would be ac^3t^3 , and the fifth term would be ac^4t^4 . Look more closely at the pattern, and it will reveal the a stays a but the c and d powers increase with the number of terms. The term number is always one greater than the exponents on c and t . Therefore, the 55th term has 54 as exponents for c and t , but a stays the same.

$ac^{54}t^{54}$ Answer D Level 9

12) To answer this question correctly, use the n^{th} term of an arithmetic sequence formula to set up two different equations from the given information. $A_n = A_1 + (n - 1)d$: $22 = A_1 + (5 - 1)d$ or $22 = A_1 + (4)d$, and the second equation $172 = A_1 + (55 - 1)d$ or $172 = A_1 + (54)d$.

Remove parenthesis and put them into standard form to set up elimination:

$$A_1 + 4d = 22$$

$$A_1 + 54d = 172$$

Multiply the top row by -1 and add the top row to the bottom row.

$$-A_1 - 4d = -22$$

$$A_1 + 54d = 172$$

$50d = 150$, $d = 3$. Now substitute d in either original equation to solve:

$A_1 + 4(3) = 22$, $A_1 = 10$. Now find the 105th term of the sequence using the n^{th} term of an arithmetic sequence: $A_{105} = 10 + (105 - 1)3 = 10 + (104)3 = 322$

322 Answer B Level 8.5

13) To find the n^{th} term of any arithmetic sequence, use the formula:

$A_n = A_1 + (n - 1)d$. This sequence starts at 2 and has a common difference of 2, so the 177th term is $A_{177} = 2 + (177 - 1)2 = 354$

354 Answer A Level 4